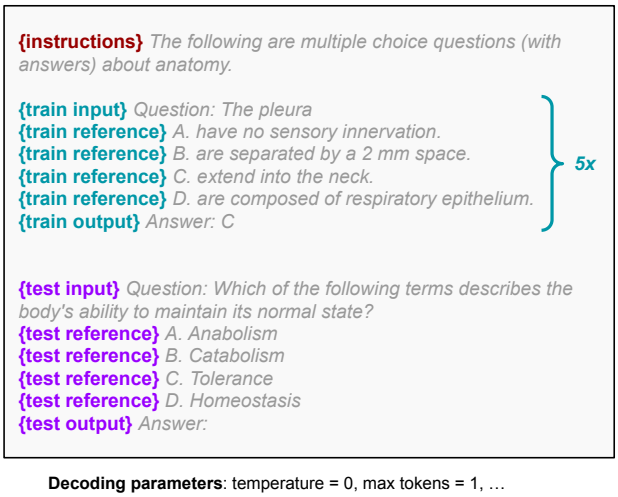

Figure 23: **Prompt formatting.** An example of how we structure and format the prompt for querying the language model.

	Parameter	Language Modeling	**TruthfulQA**	**CNN/DailyMail**
Prompt format §J.1: PROMPTING-TEST §J.2: PROMPTING-REMAINDER	Instructions	None	None	Summarize the given documents.
	Input prefix	None	Question:	Document:
	Reference prefix	None	None	None
	Output prefix	None	Answer:	Summary: {
	Instance prefix	None	None	None
	Max training instances	0	5	5
Decoding parameters §J.3: DECODING-PARAMETERS	Temperature	0	0	0.3
	Max tokens	0	5	128
	Stop sequence(s)	None	\n	}
	Num. outputs	0	1	1
Evaluation parameters	Num. runs	3	3	3
	Max evaluation instances	1000	1000	1000

Table 7: **Adaptation parameters.** Example of the adaptation parameters specified for (i) language modeling scenarios, (ii) the **TruthfulQA** question answering scenario, and (iii) the **CNN/DailyMail** summarization scenario. We also include additional parameters required to fully specify evaluation (i.e. the number of evaluation instances and runs, which influence the statistical validity and reliability of the results), though they are not strictly part of the adaptation process.

In-context examples. Since we include 5 in-context examples, two core decisions are why we use this number of examples and how we select the examples. We follow Brown et al. (2020) in their choice of the number, testing in §8.2: PROMPTING-ANALYSIS how the number of examples influences performance. To select examples, we sample examples to ensure class coverage (for classification coverage) in order of class frequency, so the 5 most frequent classes will be represented in-context. Critically, we choose to fix the in-context examples across all evaluation instances, in contrast to prior work (e.g. Brown et al., 2020), to more accurately perform few-shot evaluation (Perez et al., 2021). While more realistic, this also implies our model performance can be more sensitive to the random choice of in-context examples. Therefore, we re-run all experiments 3 times, only varying the randomness in in-context sample selection, to estimate this variance: in §8.2: PROMPTING-ANALYSIS, we show results can be especially high variance for some scenarios.

Prompt formatting. Beyond the in-context examples and evaluation instance, there are many other mechanistic nuances involved in formatting the exact string that submitted to the language model. Prior work has extensively shown that the design of prompts matters (Le Scao & Rush, 2021; Liu et al., 2022b),

with Bach et al. (2022) introducing a repository for prompts.[61] Since we more broadly view language models as *interfaces*, we treat prompts as user behavior rather than explicitly optimizing prompts (Shin et al., 2020; Zhou et al., 2022). Moving forward, we further expect that it may be desirable for language models to be *interoperable* meaning different models should be standardized such that they all work well for similar user prompting behavior. We provide initial evidence of how model performance varies in the exact string formatting of the inputs in §8.2: PROMPTING-ANALYSIS.

Adaptation strategy. For many scenarios (e.g. language modeling, text summarization), the way to interact with a language model may appear clear. However, in the case of multiple choice classification, each instance has several answer choices as references (usually with one marked as correct). This structure affords flexibility in how to use a language model that traditionally does not exist in classification (where a model necessarily predicts a distribution over answer choices with the prediction being the argmax).

Prior work introduces two classes of approaches to address adaptation for multiple choice scenarios. The first is the *separate* approach employed by Brown et al. (2020), where each choice is independently scored by concatenating it with the question and computing its probability according to the LM. In this approach, the probabilities may be further *calibrated* by the probability the LM assigns to the answer choice alone. The second is the *joint* approach employed by Hendrycks et al. (2021c), where all the choices are concatenated with the question to form a prompt (e.g., "`<question> A. <choice 1> B. <choice 2> Answer:`") and the LM predicts the choice index (e.g., `A` or `B`). This resembles how one might see a multiple choice question presented on an exam or test. In §8.2: PROMPTING-ANALYSIS, we vary this adaptation strategy to test the influence on performance: we find the strategy significantly influences accuracy (among other metrics), and the most accurate strategy depends on the scenario (i.e. no strategy is uniformly more accurate).

8 Experiments and results

Our evaluation offers an immense array of model predictions along with quantitative metrics for these predictions. To understand these results, we provide a web interface.[62] This interface not only provides the quantitative results, as is customary for other benchmarks and leaderboards, but also the underlying model predictions and the exact inputs and prompts that yielded these predictions. We encourage exploration of these results interactively: we believe grounding and interrogating various quantitative trends by mapping them to explicit model behaviors is necessary for the community to have common understanding of these models. Here, we provide a succinct analysis, foregrounding unique aspects of our evaluation that are made possible by its broad, holistic, and systematic nature.

8.1 Meta-analysis

To summarize, we evaluate 30 language models (e.g. davinci (175B), BLOOM (176B)) on 16 core scenarios (e.g. **NaturalQuestions**, **IMDB**) for 7 metric categories (e.g. accuracy, fairness). Here, we use our comprehensive evaluation to provide answers to important questions in the field like whether model accuracy correlates with scale or if more robust models are less biased.

Inter-metric relationships. A central tenet of our holistic evaluation is to foreground metrics beyond accuracy, ensuring that the many desiderata we should expect of systems are indeed measured for each scenario (when possible; see Table 4). As a consequence, this clarifies how different metrics relate to each other. Many prior works have explored these types of relationships extensively for specific metric pairs: for example, accuracy-robustness (Miller et al., 2021; Koh et al., 2021), accuracy-efficiency (Coleman et al., 2017), calibration-fairness (Pleiss et al., 2017; Jones et al., 2021), and bias-toxicity (Sap et al., 2019a; Halevy et al., 2021). In the context of language models, we may posit the relationships found in prior works will continue to bear, but many pairs have not been studied and we do not have definitive evidence that these relationships hold up reliably.

[61] `https://github.com/bigscience-workshop/promptsource`
[62] `https://crfm.stanford.edu/helm/v0.1.0`

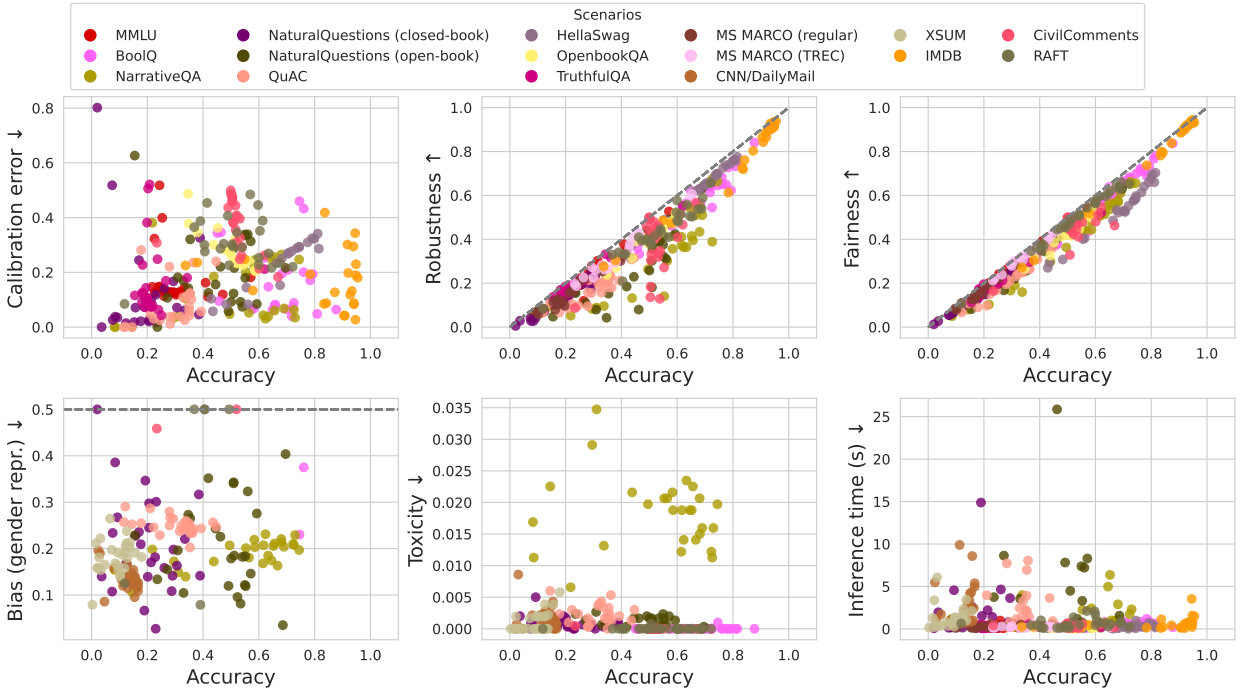

Figure 24: **Accuracy vs. X.** The relationship between accuracy (x-axis) and each of the 6 metrics (calibration, robustness, fairness, social bias, toxicity, efficiency) we study in this work across all core scenarios and for all models. For calibration error, we measure ECE-10; for bias, we measure bias in gender representation; and for efficiency, we measure denoised inference time.

Therefore, we harness our evaluation's coverage of these metrics to provide two resources. First, in Figure 24, we depict the relationship between accuracy and each of the 6 other desiderata. Among other interpretations, this figure helps clarify when more accurate systems coincide with improvements in other key desiderata and when there are important trade-offs. Further, by plotting these on a per-scenario basis, this helps to showcase the underlying heterogeneity in the trends: when are metric relationships consistent across scenarios, when are they highly scenario-dependent, and what are the anomalies? Second, in Figure 25, we consolidate the relationship across models for specific scenarios. Concretely, in each subfigure we consider a pair of metrics, where each grey point corresponds to the Pearson correlation (i.e. linear fit) between these metrics across all models for this scenario.[63] Once again, this helps to showcase the spread and heterogeneity in pairwise metric relationships as well as the macro-level trends.

While these figures together provide a wealth of information, we step through them to state specific takeaways. Most strikingly, we find that across all scenarios, accuracy, robustness, and fairness are extremely strong correlated (see Figure 25). In part, we believe this is a finding contingent on how we chose to measure robustness/fairness. For robustness, this aligns with findings for other forms of (non-adversarial) robustness, such as the work of Miller et al. (2021) for distributional robustness, which shows in-domain and out-of-domain accuracy lie on a line. On the other hand, for fairness, we find this to be quite surprising. We do emphasize several prior works instead find trade-offs between fairness and accuracy, but we do not believe that our work should be interpreted as contradicting their findings. Not only do we measure fairness differently from these works, but the setting of few-shot prompting of language models is considerably different from the settings studied in these prior works (Zhang et al., 2019a; Raghunathan et al., 2020; Dutta et al., 2020; Wang et al., 2021c; Zhao & Gordon, 2022).

[63] We also considered Spearman correlation (i.e. monotonic fit), given this may be more appropriate for some metric relationships, but found the qualitative trends to be largely invariant across these two choices of correlations metrics.